Incorporating Comment Text into Success Prediction of Crowdfunding Campaigns

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Introduction

For start-ups and small business, financing is always the key activity in order to succeed at their initial stages (Cassar 2004; Lerner 2001). To date, crowdfunding has attracted the public interest and emerged as an alternative way of external financing source for entrepreneurial start-ups to bridge the early-stage gap in raising capital from angel investors, banks, and venture capital. Unlike financing from traditional ventures, creative projects can be funded, most through the Internet, from a large number of ordinary audiences instead of a small group of specialized investors (Belleflamme et al. 2014; Mollick 2014). Similar to crowdsourcing, the concept of crowdfunding makes an open call to a large group of people to obtain the feedbacks, efforts and collective intelligence to develop interactive projects and accomplish the tasks together (Howe 2008). However, different from crowdsourcing, the ultimate objective of crowdfunding is to achieve the funding goal by collecting the monetary contributions from the crowd.

Per Crowdfunding Industry Report, crowdfunding platforms in 2015 raised around 34.4 billion dollars worldwide, a more than double increase from 12.2 billion dollars in 2014 (http://crowdexpert.com/crowdfunding -industry-statistics/). Despite the spectacular growth of the funding volume, academic research on this increasingly popularity of phenomenon is still limited. The current business models in crowdfunding sites can be distinguished into two categories based on their activities and return for the funds. One type of crowdfunding is investment-based, including equity-based, royalty-based and lending-based, by which funders are viewed as investors who expect momentary benefit in return. The other type of crowdfunding is reward-based or donation-based, where the funders receive non-monetary compensation for their financial contributions (Belleflamme et al. 2015; Kuppuswamy et al. 2015).

Past studies have observed that rewards seem to be most appealing to funders in crowdfunding sites (Gerber and Hui 2013; Hui et al. 2014). Thus, in the early stage, majority of crowdfunding platforms were donation-based, while in 2014, reward-based become the fastest growing form and has the largest number of participants, followed by donation-based and lending-based projects. Therefore, in this research we focus on reward-based crowdfunding sites, such as Kickstarter, one of the world's most prominent reward-based crowdfunding platforms, which provide tangible or intangible but non-financial rewards for their project backers. Expectably, previous studies show that rewards are viewed as the key motivations for participating in crowdfunding campaigns (Kuppuswamy et al. 2015).

Crowdfunding in Kickstarter has an essential rule "all-or-nothing", which dictates that a project must meet or exceed its fundraising goal before the clock winds down to receive the accumulated pledges during the project campaign. Kickstarter also advices that projects lasting 30 days or less tend to have higher success rates statistically. Thus, making projects successfully funded in time is very imperative to project creators, not only raising the initial funds to get the project ideas off the ground, but also gaining exposure and helping them to get attention to other potential angel investors.

Past studies have examined factors contributing to the success of crowdfunding projects (Belleflamme et al. 2014; Mollick 2014); notable factors include preparedness and social capital. Several determinants to the success of crowdfunding campaigns have been identified such as goal (the amount to be funded), duration (the period of time to raise fund), and category (the type of the project), locations(the location of the project) and social network (the relationship between project owners and backers) (Agrawal et al. 2011; Etter et al. 2013; Greenberg et al. 2013). Of the various determinants, social network has been identified as a key factor to increase funding probability (Belleflamme et al. 2015). Due to the time limit of the project (usually 4- 6 weeks), it is a good strategy to use social network influence to increase the number of backers. Agrawal et al. (2014) also indicate that social network such as friends and family plays a key role in the early stages of crowdfunding fundraising; furthermore, it also suggests that with high numbers of early contributions or backers right after the start of a project, the likelihood of funding project campaign successfully can be increased. Not surprisingly, with a solid social network, the project campaign will get off to a good start.

Nonetheless, there has been no comprehensive understanding on how project creators can make their efforts on social network community by texts so as to influence their backers and increase the success probability of their campaigns. The most common type of social text is the comment text or updates made by project creators and backers which, however, is largely ignored. In this paper, we focus on examining the influencing features of the project, especially textual features from project comments which may affect the funding success.

In this research, we propose a method that combines machine-learning analysis and text-mining techniques to identify the determinants of a successful crowdfunding project across different project's durations. We investigate two types of social information in a project campaign—backer's comments and creator's updates—and quantitatively and dynamically assess how closely the two are associated with the final success of a campaign. With this study, our major contribution on crowdfunding success is to provide more comprehensive understanding on which factors contribute to the success of crowdfunding projects. Results from our experiments show that several projects baseline features (e.g., duration, presence of a video, and goal amount) have substantial predictive power, yielding a model with high accuracy (85%). Incorporating dynamic features from backer's comments and creator's updates by different project duration further improve the accuracy of success prediction. Moreover, text-mining feature sets derived from comments and updates also contribute to the increase of prediction accuracy under various classification algorithms. This suggests that the language used by creators to pitch their project plays an important role in driving the project's success, accounting for 58.56% of the variance around success prediction.

The paper is organized as follows. In section 2, we present the related work of the crowdfunding research. We then illustrate our approach in Section 3. The experimental results with the empirical data are described in Section 4. In the last section, we conclude and point out future research directions.

Related Work

Success factors in Crowdfunding

Several previous researches have examined on this new method of financing venture. Belleflamme et al. (2014) defines crowdfunding as "an open call, mostly through the Internet, for the provision of financial resources either in the form of donation or in exchange for the future produce or some form of reward to support initiatives for specific purposes." Existing studies have identified a variety of features that may directly or indirectly influence the success of project campaign. Some researchers find that project properties , such as category, goal, duration, and with video or image are associated with funding successes (Drabløs 2015; Greenberg et al. 2013). Mollick (2014) identifies some key features on the dynamics of success and failure of Kickstarter projects, such as social capital, preparedness and geography. Previous researches also have reported that receiving substantial contributions in early stage of the project can largely predict the success of crowdfunding campaigns (Agrawal et al. 2011; Colombo et al. 2014). Additionally, communication with project creators and backers is identified to be crucial to the success of a project campaign. Hui et al. (2012) present a qualitative study of crowdfunding work by conducting interviews with project creators and participant observation to measure the work needed and how to accomplish the work for the project creators. The study finds that project creators with successfully funded projects tend to make lots of efforts on social network (on and off-line) for funds during the project campaign. Xu et al. (2014) indicate that the communications with content and updates by project creators to provide progress reports to backers also largely influence crowdfunding success. The success rate of project campaigns with updates can be reach 58.7%, while that of project campaigns without updates is only 32.6%. It clearly shows that communication plays an important role on project success.

Text-mining in Crowdfunding

Recent research in crowdfunding has gained attention in the text-mining communities. Prior works reveal that the success of a project is related to the textual content of the project. Greenberg et al. (2013) propose a model for predicting the chances of success campaigns by using some simple text-mining features from project description, such as sentiment polarity and sentence counts. Mitra and Gilbert (2014) examines the predictive power of project content, specifically the words and persuasive phrases the creator uses. Du et al. (2015) present the influence of project description in terms of

argument quality and source credibility on funding success. The numbers of updates and the types of updates are found to be positively related to the success of a project (Xu et al. 2014). However, although social capital of project creators has been identified as a key factor for campaign success, how social interaction content of a crowdfunding project can influence backers to support projects has not been examined. In our research, we hope to get more understanding on social network influence on the success chances of a project.

Machine Learning in Crowdfunding

Recent studies adopt machine learning classifiers as a tool to predict the chances of successfully funding a Kickstarter project. They employ machine learning classifiers such as Support Vector Machine (SVM) and Decision Trees (DT) to predict crowdfunding success with their accuracy rate around 76% (Etter et al. 2013; Greenberg et al. 2013). Social feature analysis on SNS such as twitter is also identified to predict the numbers of bakers and funding success (Lu et al. 2014). These previous researches offer valuable contributions and examine the determinants of successful funding and the motivating factors behind crowdfunding.

Despite the above-mentioned novel approaches, their analysis is based on very basic set of features such as number of updates, comments, Facebook friends, project description length, etc. Therefore, we build on their prior work and extend it by taking into account the social information of a project using more comprehensive text-mining technique such as sentiment analysis and text quality analysis. Additionally, our prediction model is based on the state-of-the-art gradient boosting model that achieves a much higher accuracy than the SVM classifier used in others papers. To the best of our knowledge, we are the first to utilize two types of social information—backer's comment text and creator's updates text created during project campaigns and employ gradient boosting algorithms to build an effective prediction model for Kickstarter projects.

The Approach

Baseline Features

Each crowdfunding project may have various fields, including title, creator name, category (and subcategory) to which the project belongs, project description, starting/ending date, state/country, project updates, creator/backer comments, and the final status. The final status, is the target variable to be predicted: success or failure. Drawing upon previous studies (Greenberg et al. 2013; Mollick 2014) and considering the data available from Kickstarter, we choose or formulate 11 features to be our baseline set of features, as shown in Table 1. Note that we distinguish two types of features: static and dynamic. Static features are those whose values never change throughout the campaign. Dynamic features, comprising counts of updates and counts of comments by project duration, on the other hand, have values constantly changing since the start of campaigns. Updates represent efforts by project creators to reach out to potential or interested backers by informing them current project progress. Previous research shows that without early updates, the chance of success can reduced by over 13% (Mollick 2014). Comments are created by current and potential backers to ask questions or give their feedbacks. Thereby, we consider three types of intervals for a project campaign, namely first week, first two weeks, and the complete duration. An update or a comment of a project campaign can be assigned to these intervals based on the posted time of the update or comment. We also measure the number of images or videos in each project as our baseline features. In addition, the currency of each funding goal is based on creator's address; thus, we convert all other foreign currencies of funding goal into USD dollars.

Feature Name	Data Type	Feature Type	Description
category	Categorical	Static	The category to which the project belongs
subcategory	Categorical	Static	The subcategory to which the project belongs
comment_count	Int	Dynamic	The number of comments
update_count	Int	Dynamic	The times of the project description being updated
has_video	Boolean	Static	Whether the project has a video inside its description

body_length	Int	Static	The length of the project description
body_image_count	Int	Static	The number of images inside the project description
body_video_count	Int	Static	The number of videos inside the project description
goal_amount	Float	Static	The goal amount of the project in USD
duration	Int	Static	The duration of the project in days
loc_state	Categorical	Static	The location of the project founded

Table 1. Baseline Features

Features based on Text-mining

Textual information is one of the most prevalent features across all crowdfunding platforms. However, rich textual content is also difficult and time-consuming to process. Recent studies of text mining in many fields can help us extract text features. In this section, we describe text-mining features from project comments and updates, which can be divided into two categories: text sentiment and quality.

Sentiment analysis

Sentiment analysis is a prevalent technique of NLP which derives subjectivity, emotion and opinions from textual (Liu 2012; Pang and Lee 2008). For further analysis, we extract the sentiment from the content and determine the polarity of sentiment (positive or negative) on each comment by using lexicon-based sentiment analysis (Bott 2014). In this research, we adopt word sense disambiguation (WSD) and sentiment determination in NLTK package to conduct the sentiment analysis. Given a sentence, the WSD module identifies the sense (or meaning) of each word in the sentence based on WordNet. Then the sentiment of the identified sense is determined by referring to SentiWordNet, an expansion of WordNet by attaching sentiments to senses. The output of sentiment grading contains three scores, indicating the polarity probabilities of neutral, positive and negative. The sentiment of a comment is the average sentiment of its constituent words. The sentiment of a project is, in turn, the average sentiment of its comprising comments. In addition, three kinds of verbs are also involved: stative verbs, interpreted action verbs and descriptive action verbs. A stative verb refers to a verb that represents a long-term action (e.g., think, see and hear). The list of stative verbs was recorded as a dictionary in advance (Lakoff, George 1966). Next, we calculate the percentage of stative verbs occurred in each comment to be a feature. Furthermore, we also select our features on the percentage of interpreted action verbs with sentiment polarity (e.g., hate, love, enjoy, entertain) and descriptive action verbs (e.g., speak, say, write), which are verbs with neutral or without sentiment polarity.

Text Quality Analysis

(Liu et al. 2008) identify several indicators for text quality, namely sentence level informativeness, word level informativeness, and feature level informativeness, where sentence level informativeness refers to the number of sentences, the average length of each sentence and the number of sentences with desired feature. Additionally, some previous studies also verify the quality of text by classifying features from textual content based on the readability of the text, the reputation of the writers, and various content features based on the content terms (Ghose and Ipeirotis 2010). Several statistics of text structure are also measured in our feature set, such as the average number of sentences/words in a comment and the average count of words in a sentence. Following common practice in existing literature that examines the role of texts in contexts, we determine to focus on linguistic features of the text, or how texts are written. Thereby, based on Lakoff, George (1966), we adopt the percentage of descriptive words (e.g., adjectives and adverbs) to our feature set to measure the extent to which project backers or creators explain their feelings and experiences. To enhance our text quality analysis, we also consider readability. Readability of text is to measure how accessible the texts are. Existing research in many fields has proved that readability is a simple but very effective indicator for the writer's capabilities (Gao and Lin 2016). We adopt the popular readability indicator, the Automated Readability Index (ARI), which is used for estimating how easily the text to be read, and whose formula is as follows.

ARI = 0.50 (words per sentence) + 4.71 (strokes per word) – 21.43..... (1)

As a result, we develop 13 text-mining features listed in Table 2.

Feature Name	Data Type	Description	Sentiment/Quality
IAV	float	Average percentage of Interpretive Action Verbs in comments	Sentiment
DAV	float	Average percentage of Descriptive Action Verbs in comments	Sentiment
posiCounter	float	Average positive sentiment scores in comments	Sentiment
negaCounter	float	Average negative sentiment scores in comments	Sentiment
openClassPOS	float	Average percentage of parsed tokens that are open-class in comments	Text Quality
sentenceOverallAVG	float	Average number of words in a sentence	Text Quality
adjAdv	float	Average percentage of adj and adv in a sentence	Text Quality
sentenceNum	float	Average number of sentences in a comment	Text Quality
stativeV	float	Average percentage of stative verbs in a sentence	Text Quality
ARI	float	Average automated readability index of the comments	Text readability
adj	float	Average of percentage of adj in a sentence	Text Quality
wordsNum	float	Average number of word in a comment	Text Quality
characterPerWord	float	Average ord length	Text Quality

Table 2. Text-mining Features

Feature Selection

Feature selection is a critical task for classification, whose goal is to identify those features in the data that are most useful to predict the desired outcome. To automatically determine our relative importance of each features listed in Table 1 and Table 2, we apply boosting models to decide what features should we use to create our predictive model. Boosting Models automatically provides estimates of feature importance from a trained predictive model, where each split search for the best feature and splitting point to optimize the objective outcome (Schapire 2003). After calculating the gain on each node and measuring the contribution of the selected features, we verify all the trees, and sum up all the contribution by information gain score for each feature and treat it as the importance (Chen and Guestrin 2016).

Predictive model

To predict the success of a crowdfunding project, we need to build the predictive model. In our work, a boosting machine learning method is applied to construct the prediction model. Boosting is a general approach that is used to improve the performance of any machine-learning models, including regression and classification algorithms. The basic idea of boosting is to gradually reduce the error in every iteration, and of the various morphs of boosting machines, the definition of error and ways to reduce it differ. For example, Adaboost fits an ensemble model in a forward stage-wise manner, which means in each iteration, the machine introduces a weak learner on the data and try to label the misclassified data in the previous stage with correct class (Freund and Schapire 1995). In Adaboost, the shortcomings to be minimized are identified by high-weight data points, which are the aforementioned misclassified data. The gradient boosting machine shares the same concept as Adaboost that correct the mistake throughout the iterations and further combines the gradient

descent and boosting in order to minimize the error function by moving in the opposite direction of the gradient (Friedman 2011) . In our work, we use the eXtreme Gradient Boosting (XGBoost), which extends the gradient boosting and strengthen its ability in sampling and multi-threaded process, as our boosting algorithm (Chen and Guestrin 2016).

Empirical Evaluation

Data collection

We use a pre-scraped dataset of project pages provided by CrowdBerkeley (https://crowdfunding.haas.berkeley.edu), a research institute sponsored by UC Berkeley's Fung Institute and the Kauffman Foundation that provides data of several crowdfunding sites in open access. Specifically, we employ a list of 220,524 Kickstarter projects since the launch of Kickstarter on June 2, 2009. Kickstarter has become the largest and dominant crowdfunding site in the world. According to the website of Kickstarter, their projects are grouped into 15 categories: Art, Comics, Dance, Fashion, Design, Film, and Video, food, Games, Music, Photography, Publishing, Technology and Theater. For predicting the final status of the campaigns, we include only those projects which have reached their end date and thus have a clear outcome: Success or Failure. In addition, we remove those projects that have no comments and updates in order to evaluate the effectiveness of our proposed text-mining based features. Thus, the resultant dataset contains 88,211 projects with 7,478,720 comments, as shown in Table3 .

Dataset Statistics	
# of projects	88,211
# of successful	60,200
# of Failed	28,011
# of category	15
# of comments	7,478,720

Table 3. Dataset Statistics

. Experiment Design

To evaluate the impact of time duration on prediction accuracy, we assign comments and updates into three types of time intervals: first one week, first two weeks, and the complete duration. We then construct a prediction model for the data set in each time window. We apply NLTK package on the data set to determine feature values. First, we run the tokenize module of NLTK to segment the comment or updates text into words, meanwhile, we adopt the built-in stopwords module and a manually built list to remove the stopwords, punctuation, and numbers. Next, the POS module is conducted to label the POS tag of each token, and lemmatize the tokens to return to the base form of the word. The stemming module is applied to find the word root. After the processing, we are able to determine the values for both baseline features and text-mining features, which are used to construct the prediction model.

. Evaluation Result

To identify the important features from our feature sets in Table 1 and Table2, we apply Xgboost package in Python to rank the importance of all features. Figure 1 shows the corresponding percentage of relative importance over the feature sets. Although the importance of our text-mining features is generally lower than that of the baseline features, some of them (e.g., ari, sentenceOverAvg, and openClassPOS) are still shown to be relatively significant for success prediction.

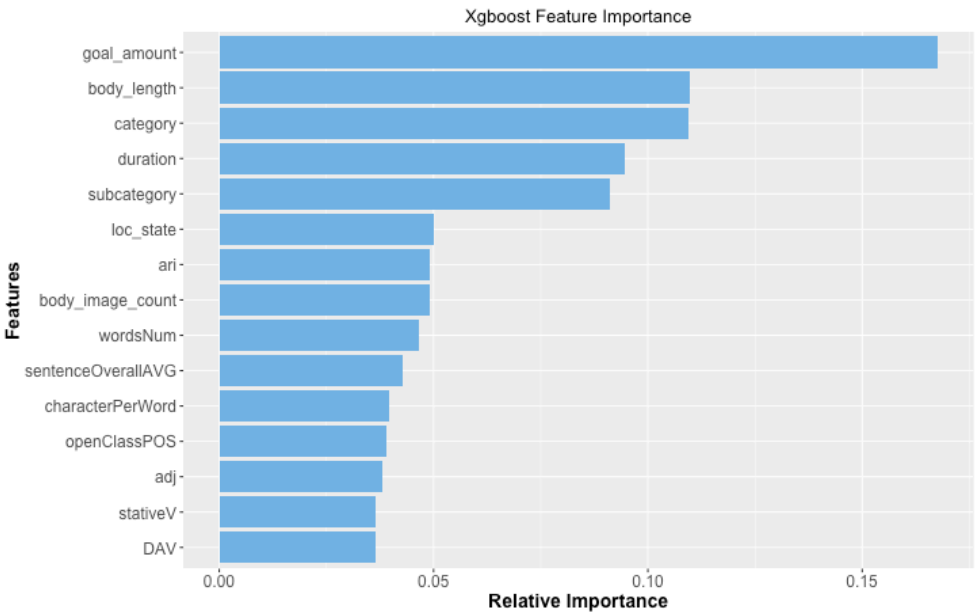

Figure 1. XgBoost Relative Importance of Features

Figure1 shows that the funding goal and category are key factors to the success of projects, which is consistent to the findings in previous studies (Etter et al. 2013; Mollick 2014). See Figure2.

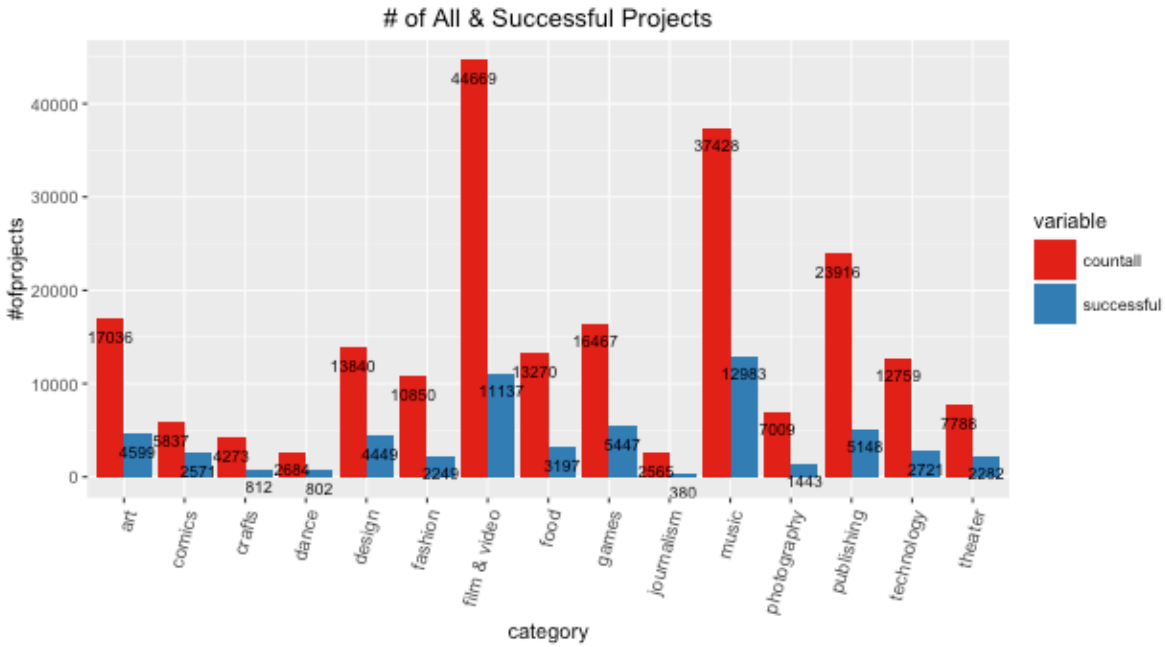

Figure 2. # of projects in each category

Prediction of Success We build the prediction models using three different feature sets based on data from three different time windows. The three feature sets are static features, baseline features (i.e., static and dynamic features) and complete features (baseline features and comment text analysis features). Four premium classification algorithms, namely GLM, Random Forest, Gradient Boosting and XgBoost, are employed with 10-fold cross validation. The result is shown in Table 4, 5, and 6. As can be seen, the complete features achieve the best performance across different classifiers, followed by Baseline with dynamic features, though the difference is not much. Comparing the accuracy in Table 4, 5, and 6 with those in the previous subsections, we can say that using the text-mining features further improves the classification accuracy. As the length of time window increases, the accuracy of the constructed prediction model also increase. Finally, as expected, of the various classification algorithms, the ensemble method XgBoost achieves the best performance.

Algorithms	W1 Baseline Feature						Baseline Features and Text-mining Features		
	Static features			Static + Dynamic features					
	Acc	Pre	Rec	Acc	Pre	Rec	Acc	Pre	Rec
GLM	75.8%	77.1%	90.6%	76.4%	78.2%	91.2%	77.2%	78.5%	92.9%
Deep Learning	81.8%	82.2%	92.3%	82.3%	82.7%	93.3%	83.1%	84.0%	93.5%
Gradient Boosting	82.5%	82.3%	93.1%	83.2%	83.5%	93.2%	83.6%	84.2%	94.2%
Ensemble	82.9%	83.1%	93.3%	83.1%	83.7%	93.8%	83.5%	83.7%	94.1%
XgBoost	84.6%	85.5%	93.3%	84.8%	85.7%	93.4%	85.2%	85.9%	93.5%

Table 4. Projects with comments in the first one week

Algorithms	W2 Baseline Feature						Baseline Features and		
	Static features			Static + Dynamic features			Text-mining Features		
	Acc	Pre	Rec	Acc	Pre	Rec	Acc	Pre	Rec
GLM	76.1%	76.9%	92.5%	76.5%	78.5%	91.7%	77.7%	78.8%	92.3%
Deep Learning	82.7%	83.2%	93.3%	83.1%	83.7%	93.5%	83.1%	83.6%	93.5%
Gradient Boosting	83.1%	83.6%	93.7%	83.3%	83.8%	93.7%	83.6%	84.0%	94.0%
Ensemble	83.3%	83.5%	92.9%	83.5%	83.7%	93.8%	83.6%	83.7%	94.1%
XgBoost	84.6%	85.5%	93.3%	85.8%	84.9%	93.3%	86.1%	85.3%	93.6%

Table 5. Projects with comments in the first two week

Algorithms	Complete Baseline Feature						Baseline Features with		
	Static features			Static + Dynamic features			Text-mining Features		
	Acc	Pre	Rec	Acc	Pre	Rec	Acc	Pre	Rec
GLM	76.8%	78.2%	91.7%	82.3%	83.4%	92.5%	82.9%	83.4%	92.6%
Deep Learning	76.8%	78.2%	91.7%	82.3%	83.4%	92.5%	82.9%	83.4%	92.6%
Gradient Boosting	82.8%	83.1%	93.8%	85.8%	86.4%	94.0%	86.2%	87.3%	93.4%
Ensemble	83.4%	83.8%	93.8%	85.8%	86.3%	94.2%	86.3%	86.6%	94.6%
XgBoost	83.5%	84.6%	92.7%	86.5%	87.5%	93.5%	86.8%	87.7%	93.8%

Table 6. Projects with comments in the complete duration

Conclusion

In this paper, we intend to identify the set of features for predicting a crowdfunding project's success. Thereby, we extend previous knowledge to determine the most important features for predicting crowdfunding success. While prior studies have identified several critical features to funding success, little research is conducted explore the social information content features (such as backer's comment an creator's updates).

In our work, we use a completed real-world dataset of Kickstarter campaigns. We construct a success prediction model to identify discriminatory features that influence funding success, based on three kinds of feature sets: the static features, dynamic features and text-mining features. Drawing upon a large dataset of successfully and unsuccessfully funded projects of the kickstrate.com, we empirically

evaluate our research model with three features sets in three different time interval (first week, first two weeks and complete duration). In our proposed model, the experimental result reveals that the model incorporating dynamic features from backer's comment and creator's update counts accumulated during different project duration can improve the accuracy of success prediction. It also indicates early updates and comments are very critical to the funding success. With sentiment and textual quality feature set derived from social interactive comments and updates, we can further increase classification accuracy. Of the text-mining features, we find some features that significantly influence the outcome of a project, such as Stative V (e.g., hear, see, know), ari (text quality), and adj (emotional words). It also represents that project creators or backer with more emotional words could contribute to the success of project.

In summary, our results provide a promising step towards inferring the impacts of social information on crowdfunding. In the future, besides employing text-analysis method, we hope to consider the semantics of the text by looking at their discourse structures. Moreover, with the syntax rules, we could construct the sentiment lexicon in each category of the project with their own corpus to improve accuracy on sentiment analysis and discover more linguistic features.

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